

QUESTION BANK

Topic: Partial Differentiation & Applications

Q1. If $z = f(x + ay) + g(x - ay)$, then

A. $z_{xx} = z_{yy}$

B. $z_{xx} = a^2 z_{yy}$

C. $z_{yy} = a^2 z_{xx}$

D. $z_{xx} + a^2 z_{yy} = 0$

Q2. If $x = \log(x \tan^{-1} y)$, then f_{xy} is equal to

A. $\frac{-1}{x^2}$

B. 0

C. $\frac{1}{x^2}$

D. none of these

Q3. If $z = \cos\left(\frac{x}{y}\right) + \sin\left(\frac{x}{y}\right)$, then $x z_x + y z_y$ is equal to

A. z

B. $2z$

C. 0

D. $4z$

Q4. If $u = \log(x^3 + y^3 + z^3 - 3xyz)$, then $x u_x + y u_y + z u_z$ is equal to

A. $3u$

B. $2u$

C. 0

D. 3

Q5. If $u = x^2 + y^2 + z^2$ be such that $x u_x + y u_y + z u_z = \lambda u$, then λ is equal to

- A. 1
- B. 2
- C. 0
- D. none of above

Q6. If $f(x, y, z) = 0$, then the value of $\frac{\partial x}{\partial y} \cdot \frac{\partial y}{\partial z} \cdot \frac{\partial z}{\partial x}$ is

- A. 1
- B. -1
- C. 0
- D. none of these

Q7. If $u(x, y) = x^2 \tan^{-1}\left(\frac{y}{x}\right) - y^2 \tan^{-1}\left(\frac{x}{y}\right)$, $x > 0, y > 0$,

then $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2}$ is equal to

- A. 0
- B. u
- C. 2u
- D. 3u

Q8. If $f(x, y) = e^{xy^2}$, the total differential of the function at the point (1, 2) is

- A. $e(dx + dy)$
- B. $e^4(dx + dy)$
- C. $e^4(4dx + dy)$
- D. $4e^4(dx + dy)$

Q9. If $f(x, y) = 0$, then $\frac{dy}{dx}$ is equal to

- A. $\frac{f_x}{f_y}$

B. $-\frac{f_x}{f_y}$

C. $\frac{f_y}{f_x}$

D. $-\frac{f_y}{f_x}$

Q10. The function $f(x, y) = 2x^2 + 2xy - y^3$ has

A. only one stationary point at $(0, 0)$.

B. two stationary points at $(0, 0)$ and $(\frac{1}{6}, -\frac{1}{3})$.

C. two stationary points at $(0, 0)$ and $(1, -1)$.

D. no stationary points.

Q11. If $z = f(x, y)$, then dz is equal to

A. $f_x dx + f_y dy$

B. $f_y dx + f_x dy$

C. $f_x dx - f_y dy$

D. $f_y dx - f_x dy$

Q12. The function $z = 5xy - 4x^2 + y^2 - 2x + y + 5$ has at $x = \frac{1}{41}$ and $y = \frac{5}{41}$

A. maxima

B. saddle point

C. minima

D. none of above

Q13. If $f(x, y)$ is such that $f_x = e^x \cos y$ and $f_y = e^x \sin y$, then which of the following is true?

A. $f(x, y) = e^{x+y} \sin(x+y)$

B. $f(x, y) = e^x \sin(x+y)$

C. $f(x, y)$ does not exist

D. none of above

Q14. With usual notations, the properties of maximum and minima under various conditions are

I

II

(P) Maxima

(i) $rt - s^2 = 0$

(Q) Minima

(ii) $rt - s^2 < 0$

(R) Saddle Point

(iii) $rt - s^2 > 0, r > 0$

(S) Case of failure

(iv) $rt - s^2 > 0, r < 0$

A. P – i, Q-iii, R – iv, S- ii

B. P – ii, Q-i, R – iii, S- iv

C. P – iii, Q-iv, R – ii, S- i

D. P – iv, Q-iii, R – ii, S- i

Q15. If $u = f(y + ax) + g(y - ax)$, then $u_{xx} - a^2 u_{yy}$ is equal to

A. 0

B. a^2

C. $a^2(f'' - g'')$

D. $a^2(f'' + g'')$

Q16. The critical points of the function $x^3 + y^3 - 3axy = 0, a > 0$ are

A. (0, 0), (a, a)

B. (0,0), $\left(\frac{a}{2}, \frac{a}{2}\right)$

C. (0,0), $\left(\frac{a}{3}, \frac{a}{3}\right)$

D. (0,0), $\left(\frac{3a}{2}, \frac{3a}{2}\right)$

Q17. The extreme value of the function $f(x, y) = xy - x^2 - y^2 - 2x - 2y + 4$ is:

A. 0

B. 8

C. -8

D. 4

Q18. The critical points of the function $xy(a-x-y)$, $a>0$ are:

A. $(0, 0), (a, a)$

B. $(0,0), (a, 0), (0, a), \left(\frac{a}{2}, \frac{a}{2}\right)$

C. $(0,0), (a, 0), (0, a), \left(\frac{a}{3}, \frac{a}{3}\right)$

D. $(0,0), \left(\frac{3a}{2}, \frac{3a}{2}\right)$

Q19. Which of the following point/points behave as saddle point for the function

$$f(x, y) = x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x - 90$$

A. $(4, 0), (6, 0)$

B. $(0,4), (0, 6)$

C. $(1, 5), (-1,5)$

D. $(5,1), (5, -1)$

Q20. The function $f(x, y)$ has a relative maximum at point (a, b) if for small positive or negative values of h and k we have

A. $f(a, b) < f(a+h, b+k)$

B. $f(a,b) > f(a+h, b+k)$

C. $f(a, b) = f(a+h, b+k)$

D. none of above

Q21. Which of the following statement is not true:

A. Extremum is a point which is either a maximum or minimum.

B. Saddle point is a point, where the function is neither maximum nor minimum.

C. Every stationary point is an extrema.

D. Extrema occurs only at stationay points.

Q22. For the function $f(x, y) = 1 - x^2 - y^2$, the point $(0, 0)$ is a

- A. saddle point
- B. maximum point
- C. minimum point
- D. none of above

Q23. The critical point of the function $f(x, y) = x^2 + y^2 - 9a$, $a > 0$ is

- A. (0, 0)
- B. (a, 0)
- C. (0, a)
- D. (a, a)

Q24. For the function $f(x, y) = x^3 + y^3 - 3axy$,

- A. the point (a, a) is a point of maximum, if $a > 0$.
- B. the point (a, a) is a point of maximum, if $a < 0$.
- C. the point (a, a) is a point of minimum, if $a < 0$.
- D. the point (a, a) is a saddle point.

Q25. For the function $f(x, y) = xy(a - x - y)$,

- A. the point $\left(\frac{a}{3}, \frac{a}{3}\right)$ is a point of maximum if $a > 0$.
- B. the point $\left(\frac{a}{3}, \frac{a}{3}\right)$ is a point of maximum if $a < 0$.
- C. the point $\left(\frac{a}{3}, \frac{a}{3}\right)$ is a point of minimum if $a > 0$.
- D. the point $\left(\frac{a}{3}, \frac{a}{3}\right)$ is a saddle point.

Q26. The maximum value of the function $f(x, y, z) = x^2 + y^2 + z^2$, where $lx + my + nz = p$ is:

- A. $\frac{l^2 + m^2 + n^2}{p^2}$
- B. $\frac{p^2}{l^2 + m^2 + n^2}$
- C. $\frac{3p^2}{l^2 + m^2 + n^2}$

D. $\frac{l^2+m^2+n^2}{3p^2}$

Q27. The minimum value of the function $f(x, y, z) = x^2 + y^2 + z^2$ subjected to the condition $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 1$ is

- A. 18
- B. 81
- C. 243
- D. 27

Q28. The shortest distance from the origin to the surface $xyz^2 = 2$ is:

- A. 2
- B. 1
- C. 4
- D. 8

Q29. Which of the following statement is not true:

- A. The tangent planes to the surface at the maximum or minimum point is parallel to xy-plane and perpendicular to x-axis.
- B. The hyperbolic paraboloid, $z-xy=0$, has a saddle point at the point (1, 1).
- C. At a saddle point the function is maximum in one direction, while minimum in other direction.
- D. Extrema occurs only at stationary points.

Q30. Critical points of the function $f(x, y) = x^3y^2(a-x-y)$, $a < 0$ are

- A. $(0, 0), \left(\frac{a}{3}, \frac{a}{3}\right)$
- B. $(0, 0), \left(\frac{a}{2}, \frac{a}{3}\right)$
- C. $(0, 0), \left(\frac{a}{3}, \frac{a}{2}\right)$
- D. $(0, 0), \left(\frac{a}{2}, \frac{a}{2}\right)$

Q31. If $u = x^y$, the values of $\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}$ are :

- A. $xy^{x-1}, y^x \log y$
- B. $x^y \log x, yx^{y-1}$
- C. $yx^{y-1}, y^x \log y$
- D. $yx^{y-1}, x^y \log y$

Q32. If $u = f(r, s, t), r = \frac{x}{y}, s = \frac{y}{z}, t = \frac{z}{x}$ then $xu_x + yu_y + zu_z$ is :

- A. 0
- B. 1
- C. 2
- D. 3

Q33. If $x + y + z = \log z$, then z_x is:

- A. $\frac{z}{1-z}$
- B. $\frac{z}{1+z}$
- C. $\frac{1-z}{z}$
- D. $\frac{1+z}{z}$

Q34. Linear Taylor series polynomial approximation to function $f(x, y) = x^2y - 2$ about point (1,-2) is :

- A. $y - 4x + 2$
- B. $-4x + y$
- C. $y + 4x + 2$
- D. $y - 4x - 5$

Q35. If $x = au, y = bv, z = cw$, then Jacobian of (x, y, z) w.r.t (u, v, w)

- A. abc
- B. $a^2b^2c^2$

C. $abcuvw$

D. $3abc$

Q36. If $u = x^2 f\left(\frac{y}{x}\right)$ then:

A. $x \frac{\partial u}{\partial x} - y \frac{\partial u}{\partial y} = 0.$

B. $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 0.$

C. $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 1.$

D. $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2u.$

Q37. If $x = r \cos \varphi \sin \theta$, $y = r \sin \varphi \sin \theta$, $z = r \cos \theta$, then the value of $\frac{\partial(x,y,z)}{\partial(r,\theta,\varphi)}$ is :

A. 0

B. r

C. $r^2 \sin \theta$

D. $r^2 \cos \theta$

Q38. The Jacobian $\frac{\partial(u,v)}{\partial(x,y)}$ for the function $u = e^x \sin y$, $v = x + \log \sin y$ is

A. 1

B. $\sin x \sin y - x y \cos x \cos y$

C. 0

D. $\frac{e^x}{x}$

Q39. If the function u, v, w of three variables x, y, z are not independent, then Jacobian of u, v, w w.r.t. x, y, z is always equal to

A. 1

B. 0

C. ∞

D. Jacobian of x, y, z w.r.t. u, v, w .

Q40. If $u = \log(x^2 + y^2 + z^2)$, then the value of $x u_x + y u_y + z u_z$ is equal to

- A. 0
- B. $2u$
- C. 2
- D. $2e^u$

Q41. If $u = \log x + \log y$, then the value of $x u_x + y u_y$ is

- A. u
- B. $2u$
- C. 1
- D. 0

Q42. If $\theta = t^n e^{-\frac{r^2}{4t}}$ such that $\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \theta}{\partial r} \right) = \frac{\partial \theta}{\partial t}$, the value of n is

- A. $\frac{3}{2}$
- B. $-\frac{3}{2}$
- C. $\frac{1}{2}$
- D. 1

Q43. The stationary points of $x^4 + y^4 - 2x^2 + 4xy - 2y^2$ are:

- A. $(0,0), (\sqrt{2}, -\sqrt{2}), (-\sqrt{2}, \sqrt{2})$
- B. $(\sqrt{2}, -\sqrt{2}), (-\sqrt{2}, \sqrt{2})$
- C. $(0,0), (-\sqrt{2}, \sqrt{2})$
- D. $(0,0), (\sqrt{2}, -\sqrt{2})$

Q44. The plane $x = 1$ intersects the paraboloid $z = x^2 + y^2$, is a parabola. The slope of the tangent to the parabola at $(1,2,5)$ is:

- A. 4

B. 2

C. 8

D. 16

Q45. If $f(x, y) = (50 - x^2 - y^2)^{\frac{1}{2}}$, the approximate value of $f(3, 4) - f(3.1, 3.9)$ is:

A. -0.02

B. 0.02

C. 0.04

D. -0.04

Q46. If $x = r \cos \theta$, $y = r \sin \theta$, $z = z$, then the value of $\frac{\partial(x, y, z)}{\partial(r, \theta, z)}$ is

A. θ

B. r

C. 0

D. $r^2 \sin \theta$

Q47. If $u = \sin^{-1} \sqrt{x - y}$ where $x = 3t$, $y = 4t^3$, then $\frac{du}{dt}$ is:

A. $\sqrt{1 - t^2}$

B. $\frac{3}{\sqrt[3]{1 - t^2}}$

C. $\frac{3}{\sqrt{1 - t^2}}$

D. $3\sqrt{1 - t^2}$

Q48. If u is a homogeneous function of order n , then $\frac{\partial u}{\partial x}$ and $\frac{\partial u}{\partial y}$, both are homogeneous functions of the order:

A. n

B. $n - 1$

C. $n + 1$

D. $n \pm 1$

Q49. If $u = \sin^{-1} \frac{x+y}{\sqrt{x}+\sqrt{y}}$, then $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}$ is

A. $\frac{1}{2} \tan u$

B. $-\frac{1}{2} \tan u$

C. $\frac{1}{4} \tan u$

D. $-\frac{1}{4} \tan u$

Q50. The equation of normal line to the surface $xyz = a^3$ at the point (x_1, y_1, z_1) .is

A. $\frac{x-x_1}{y_1 z_1} = \frac{y-y_1}{z_1 x_1} = \frac{z-z_1}{x_1 y_1}$

B. $-\frac{x-x_1}{y_1 z_1} = \frac{y-y_1}{z_1 x_1} = -\frac{z-z_1}{x_1 y_1}$

C. $\frac{x-x_1}{y_1 z_1} = -\frac{y-y_1}{z_1 x_1} = -\frac{z-z_1}{x_1 y_1}$

D. $-\frac{x-x_1}{y_1 z_1} = -\frac{y-y_1}{z_1 x_1} = \frac{z-z_1}{x_1 y_1}$

Q1. C

Q2. B

Q3.C

Q4. D

Q5. B

Q6. B

Q7. C

Q8. D

Q9. B

Q10. B

Q11. A

Q12. B

Q13. C

Q14. D

Q15. A

Q16. A

Q17. B

Q18. C

Q19. D

Q20. B

Q21. C

Q22. B

Q23. A

Q24. B

Q25. A

Q26. B

Q27. D

Q28. A

Q29. B

Q30. B

Q31. D

Q32. A

Q33. A

Q34. A

Q35. A

Q36. D

Q37. C

Q38. C

Q39. B

Q40. C

Q41. C

Q42. B

Q43. A

Q44. A

Q45. A

Q46. B

Q47. C

Q48. B

Q49. A

Q50. A